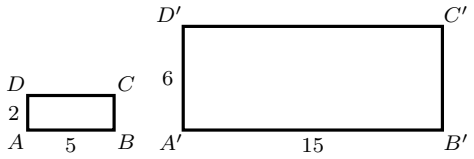
Artifact 7 – Area and Perimeter of Dilations

Driving Question: Why does area grow faster than perimeter?

Example 1

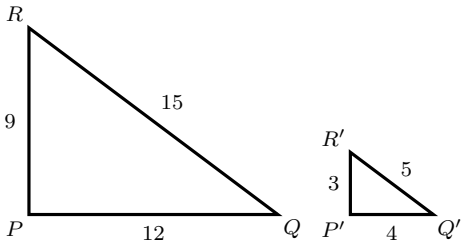
Rectangle $ABCD$ is dilated by a scale factor of 3 to create rectangle $A'B'C'D'$. Lengths are in units.



	Pre-image	Image	Image ÷ pre-image
Side			
Perimeter			
Area			

Example 2

Triangle PQR is dilated by a scale factor of $\frac{1}{3}$ to create triangle $P'Q'R'$. Lengths are in units.



	Pre-image	Image	Image ÷ pre-image
Side			
Perimeter			
Area			

Vocabulary

Perimeter
Area
Expression
Scale model

Notes to Forgetful Self

Scale factor	Perimeter ratio	Area ratio
$k = 2$		
$k = 3$		
$k = 4$		
$k = \frac{1}{2}$		
$k = \frac{1}{3}$		
k (any)		

A figure with perimeter P and area A is dilated by k.

Perimeter of the image =

Area of the image =

Try it

1. A parking lot and a scale drawing of the lot are similar rectangles. Each dimension of the parking lot is 2.4 times the corresponding dimension of the drawing. Which statement is true?

(A) The area of the parking lot is 4.8 times the area of the drawing.
(B) The area of the parking lot is 5.76 times the area of the drawing.
(C) The perimeter of the parking lot is 4.8 times the perimeter of the drawing.
(D) The perimeter of the parking lot is 5.76 times the perimeter of the drawing.

2. Polygon F has an area of $12m$ square feet. Polygon F is dilated by a scale factor of 4 to create polygon G. What is the area, in square feet, of polygon G?

(A) $48m$ square feet
(B) $3m$ square feet
(C) $192m$ square feet
(D) $768m$ square feet